



REAL ESTATE FINANCE AND INVESTMENTS

RISKS AND OPPORTUNITIES

Chapter 10

Development Pro Forma Analysis

Celina Gardens, 250-unit spec condominiums



Figure 10.1

Celina Gardens Condominiums Development Projection							
Projected Opening Date: Start of Month 25							
6-Month Period Starting with:		Year 1		Year 2		Year 3	
		Month 1	Month 7	Month 13	Month 19	Month 25	Month 31
Land and Hard Construction Costs							
Land Acquisition		(3,050,000)					
Engineering		(20,000)	(375,000)				
Construction		(145,000)	(11,354,895)	(9,004,531)	(1,689,566)		
Soft Construction Costs							
Architect		(50,000)	(1,150,000)				
Rezoning		(120,000)	(10,000)				
Fees		(200,000)	(135,000)				
Furniture			(6,000)	(8,500)	(205,000)		
Development Overhead		(102,000)	(55,000)	(65,000)	(55,000)		
Legal		(351,000)	(251,000)	(20,000)			
Loan Interest		(851)	(54,000)	(46,840)	(55,000)		
Total Development Costs	Total	(\$4,038,851)	(\$13,390,895)	(\$9,144,871)	(\$2,004,566)	\$0	\$0
Total Net Revenues	\$37,193,355	\$0	\$0	\$0	\$0	\$18,596,678	\$18,596,677
Net Cash Flow	\$8,614,172	(\$4,038,851)	(\$13,390,895)	(\$9,144,871)	(\$2,004,566)	\$18,596,678	\$18,596,677
PV of Costs at 3.00%	(\$27,618,224)						
PV of Revenues at 7.00%	\$30,883,186						
Project NPV	\$3,264,961						

Figure 10.2

Typical Planning and Construction Schedule Lengths	
Property Type	Schedule
Warehouse	9-18 months
Garden Apartments	1-2 years
Suburban Office	18-36 months
CBD Office and Highrise Residential	2-4 years
Strip Center	18-30 months
Regional Mall	3-6 years

Two Business Phases of Development

- Phase I: the negative cash flow business
 - planning, land acquisition, design and construction
 - lower uncertainty of outcome as far as cash flow goes
 - Discount rate: ~ 2-year Treasury + small risk premium
- Phase II: the positive cash flow business
 - operating or selling out the property

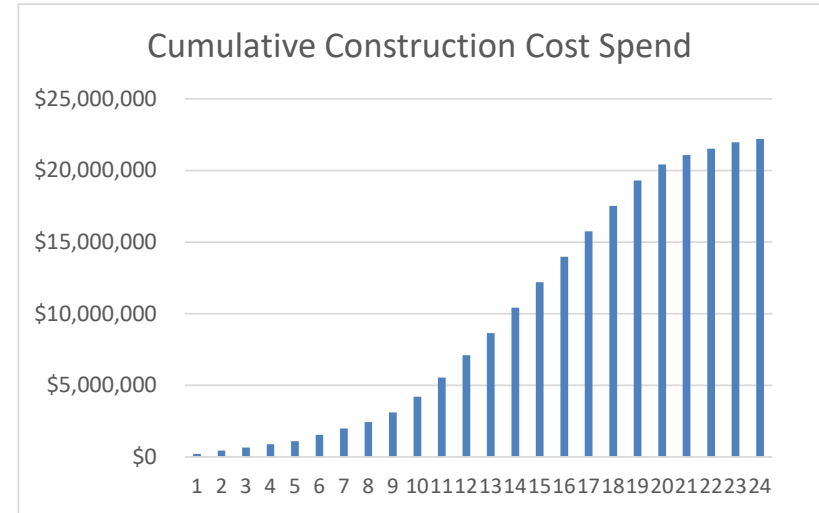
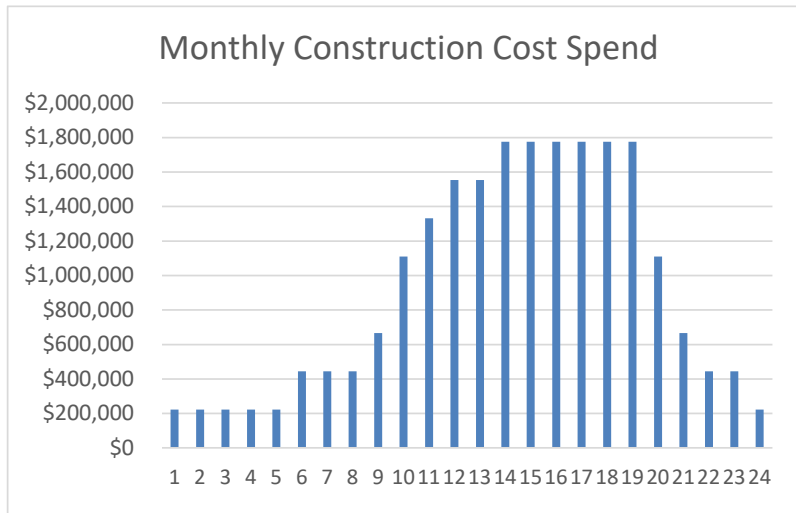
Forecasting Phase II revenues

- Difficult to forecast future NOI or property sale proceeds
- Study supply and demand for competitive space
- Pre-leasing / pre-sales (solvency risk)
- Higher discount rate should be used for positive cash flow business phase than for the negative cash flow phase

Hard Costs

- Construction contractors
 - Labor
 - Equipment and materials
 - Contractor fees
- Bell curve profile over time, accumulating into an S-curve profile

Figure 10.3



Soft Costs

- Architecture and engineering
- Project management
- Legal and accounting
- Construction loan interest
- Leasing commissions
- Developer fee

Figure 10.4

Different Quality Tenants and Discount Rate Risk						
Tenant	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
	Wish & A Prayer, Inc.			Apple		
	Rent					
	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
Appropriate Discount Rate	12.00%	12.00%	12.00%	8.00%	8.00%	8.00%

Revenue Risk Mitigation

- Pre-leasing
- Pre-sales
- Certificate of occupancy unlocks the cash stream

Delay Risk

- Causes:
 - Regulatory approvals
 - Vendor availability
 - Weather
 - Financing
 - Environmental
 - Mistakes / changes
- Impact to pre-leased / pre-sold space and financing

Opportunity

- Value creation: final product > ingredients
- Build to an “X” and sell for a “Y”
- Going-in cap rate
- Going-out (exit) cap rate
- Gross development profit margin %
 - $(\text{Expected cap rate on cost} / \text{expected cap rate on sale}) - 1$

Figure 10.6

Going-In Cap Rate* Calculation

Going-in cap rate = Projected Stabilized NOI / Projected Total Costs

Build to a 10% = \$1,000,000 / \$10,000,000

** Alternately referred to as the Yield on Cost, Return on Cost, or the Build to Return*

Figure 10.7

Going-Out Cap Rate Calculation		
Sell to an X	=	$\frac{\text{Projected Stabilized NOI}}{\text{Projected Sale Price}}$
Sell to an 8	=	$\frac{\$1,000,000}{\$12,500,000}$

Figure 10.8

Gross Development Profit Margin Calculation

Gross development profit margin = (Expected going-in cap rate / Expected going-out cap rate) – 1

(Expected going-in cap rate / Expected going-out cap rate) – 1 = (Expected value at sale / Expected cost) – 1

$$= (10 / 8) - 1 = 25\%$$

$$= (\$12,500,000 / \$10,000,000) - 1 = 25\%$$

Options and Development

- Control or ownership of developable land is like an owning an option
- Value of the option is higher the greater the growth in the local economy, the longer the hold period, and the greater the volatility of local growth

Develop vs. Buy Analysis

- Is there greater financial reward in development?
- Simulations suggest yes, but only after 15 years of cumulative cash flow

Development as a Career

- Risk tolerance is critical
- Go to where the population and job growth is